

Theory-Based Augmented User Modeling with Grounded Transformers

No Author Given

Abstract. The user model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing approaches, which represent the learner's state of mastery across specific skills or knowledge components. However, these approaches are not anchored to theoretical frameworks derived from learning theories, which limits their utility when it comes to interpreting user profiles or explaining the evolution of learning trajectories. To overcome these limitations, we propose a new approach to user modeling based on Grounded Transformers, a class of deep learning models whose internal representations are anchored to interpretable models based on cognitive or pedagogical principles. The resulting user models support instructional decisions rooted in learning theories, and we demonstrate how this grounding enables personalised, context-aware instruction in Intelligent Tutoring Systems.

Keywords: deep learning · theory-based interpretability · grounded transformers · user modeling · learning theories · intelligent tutoring systems

1 Introduction

1.1 Problem Statement

The student model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing (KT) [?] approaches, which represent the learner's state of mastery across specific skills or knowledge components. However, these traditional approaches are subject to significant limitations since they lack pedagogical interpretability, as they are not anchored to theoretical frameworks derived from cognitive or educational principles.

To overcome these limitations, Grounded Transformer have been recently proposed as a class of deep learning models able to provide information about the evolution of the user according to parameters derived from theoretical principles. In [?] they introduce gTransformer, and show how through *representational grounding* it is able to anchor its latent representations to pedagogically meaningful concepts based on Bayesian Knowledge Tracing (BKT).

In this work, we show how grounded transformers can be leveraged to create augmented student models to improve instructional decisions in Intelligent Tutoring Systems. To this end we propose gBKT, a traditional BKT model augmented with output from a gTransformer model, to enrich the student model. We evaluate gBKT, show its advantages compared with a traditional BKT model and how it can be used to create student models able to support better instructional decisions.

1.2 Research Contributions

This work makes three primary contributions:

- *Augmented Theory-Based Models*: We introduce a methodology for anchoring deep learning models to Bayesian Knowledge Tracing (BKT) logic, enabling the generation of pedagogically grounded parameters (p_{L_0} , p_T) that capture individualised student characteristics while maintaining high predictive accuracy.
- *Augmented Student Model*: We propose an augmented user model that integrates these grounded parameters to categorised learners into four distinct *learning situations* based on their prior knowledge and learning rate, providing a higher-resolution diagnostic profile than population-level models.
- *Adapted Instructional Strategies*: We demonstrate how this augmented user model enables situation-based instruction, where the ITS tutorial model can dynamically route students to differentiated instructional pathways tailored to their specific learning situation.

1.3 Research Questions

This work is guided by following research questions that together examine how Grounded Transformers [?] can be leveraged to improve theory-based augmented user modeling:

- RQ1: How can we obtain augmented student models with information derived from existent learning theories?
- RQ2: How these augmented student models can be used to improve Intelligent Tutoring Systems?

2 Background

2.1 Intelligent Tutoring Systems and User Models

Intelligent Tutoring Systems are typically structured around three interacting components:

- *domain model*: This component encodes the knowledge to be taught, typically structured as a set of Knowledge Components (KCs) or skills. It provides the semantic framework through which student performance is assessed and instructional content is organised.
- *user model*: The user model tracks the learner’s evolving state by maintaining a representation of their mastery across the domain model. In gTransformer, this is augmented with longitudinal parameters that capture the student’s unique learning patterns beyond simple correctness.
- *tutorial model*: This component mediates instructional decisions by selecting the next activity, intervention, or level of support based on the current state of the user model. The goal is to provide a personalised learning experience that optimises the student’s trajectory.

2.2 Learning Theories and Bayesian Knowledge Tracing

We select BKT [?] as our reference theoretical framework. BKT models the learning process as a Hidden Markov process governed by four interpretable parameters: Initial Mastery ($P(L_0)$), the probability that a student knows a concept prior to practice; Learn Rate ($P(T)$), the probability of transitioning from the unlearned to the learned state; Guess ($P(G)$), the probability of a correct response despite lack of mastery; and Slip ($P(S)$), the probability of an incorrect response despite mastery. These parameters enable a recursive, causally transparent estimation of student mastery that is directly interpretable by educators.

Standard BKT suffers from a significant limitation: it operates at the population level, estimating a single parameter set for all students on a given skill [?]. gTransformer addresses this by leveraging Transformer attention mechanisms to infer student-specific parameters from individual interaction histories.

2.3 Grounded Transformers

3 Theory-Based Augmented User Models

We formalise the proposed ITS architecture by integrating traditional components with our grounded modelling engine. The *domain model* defines the structured knowledge components to be mastered. The *user model* tracks student mastery across these components, and is augmented with student-specific parameters (p_{L_0}, p_T) provided by *gTransformer*.

The *gBKT* component performs Bayesian updating of student mastery L_t , where the initial belief L_0 and the learning transition T are dynamically set by gTransformer’s grounded outputs. These parameters encapsulate the student’s longitudinal learning context, allowing the user model to differentiate between learners who might show similar correctness patterns but possess different underlying learning rates or prior knowledge. The output of this augmented user model is used by: 1) the tutorial model to select instructional strategies, and 2) a diagnostic roster module that visualises student evolution in a 3D (p_{L_0}, p_T, t) space.

4 Methodology

To address RQ1, we evaluate *gBKT* by comparing its predictive performance against traditional population-level BKT. Unlike standard BKT which uses constant parameters for all students on a given skill, gBKT leverages gTransformer to infer context-aware parameters from historical interaction sequences. gTransformer uses a Transformer-based encoder-decoder architecture where latent representations z_t are projected onto BKT semantic axes through representational grounding. The model is trained on benchmark datasets such as AS2009, with the grounded parameters serving both to constrain predictions to pedagogical logic and to augment the student profile.

To address RQ2, we define four *learning situations* based on the estimated (p_{L_0}, p_T) : (1) *slow starters* (low L_0 , low T), (2) *diligent beginners* (low L_0 , high T), (3) *plateaued learners* (high L_0 , low T), and (4) *fast masters* (high L_0 , high T). These situations allow the tutorial model to adapt strategies: situation (1) might receive targeted scaffolding, while (4) receives accelerated content. Since parameters are dynamic, students can transition between situations as more evidence is gathered. To illustrate this, we generate 3D roster plots showing the longitudinal trajectory of representative students across these quadrants.

5 Results and Discussion

Comparison of predictive performance (Table ??) addresses RQ1, showing that gBKT significantly outperforms standard BKT, with an average AUC gain of 19.9% ($\mathcal{I}_{\text{gain}}$). This validates that context-aware parameters derived from grounded Transformers accurately capture the nuances of individual learning journeys while remaining within the interpretable framework of BKT.

The 3D roster plots address RQ2, revealing how student profiles evolve over time. By observing the movement of students through the (p_{L_0}, p_T) space over the course of their interactions, educators gain theory-grounded insights into whether a student is disengaging (decreasing T) or gaining confidence (increasing L_0). This ability to ground deep learning state into pedagogically recognisable constructs enables more mindful instructional support and satisfies the accountability requirements for high-risk educational AI.

The parameter orbits of representative students across the four learning situations are characterised in Figures ?? and ?. Figure ?? shows covariance ellipses (1σ and 2σ) in (p_{L_0}, p_T) space, summarising the shape and extent of each student’s parameter orbit. The ellipses reveal that Diligent Beginners and Fast Masters occupy larger, spatially extended orbits—reflecting rich contextual variation across skills—while Slow Starters and Plateaued learners exhibit more compact, constrained orbits concentrated in low- and high-mastery regions respectively. The orientation and eccentricity of each ellipse indicate whether variability is predominantly along the mastery axis, the learning rate axis, or both, providing a concise geometrical signature of each learning situation.

Figure ?? shows return maps (lag plots) for the same four representatives. Each panel plots the value of a parameter at interaction $t+1$ against its value at interaction t , separately for p_{L_0} (left sub-panel) and p_T (right sub-panel), with points coloured from dark (early interactions) to bright (late interactions) using a sequential palette. The diagonal represents a fixed point: points on the diagonal indicate no change between consecutive interactions. For Slow Starters (student 600), the p_{L_0} return map shows a broad scatter around the diagonal with a slight downward shift from early (mean 0.61) to late interactions (mean 0.48), indicating persistent difficulty without recovery; p_T remains clustered near zero throughout, consistent with a low-rate attractor. Plateaued learners (student 2116) show the opposite p_{L_0} trajectory: early points scatter in mid-range values and late points consolidate near the diagonal at high mastery (0.80), while p_T

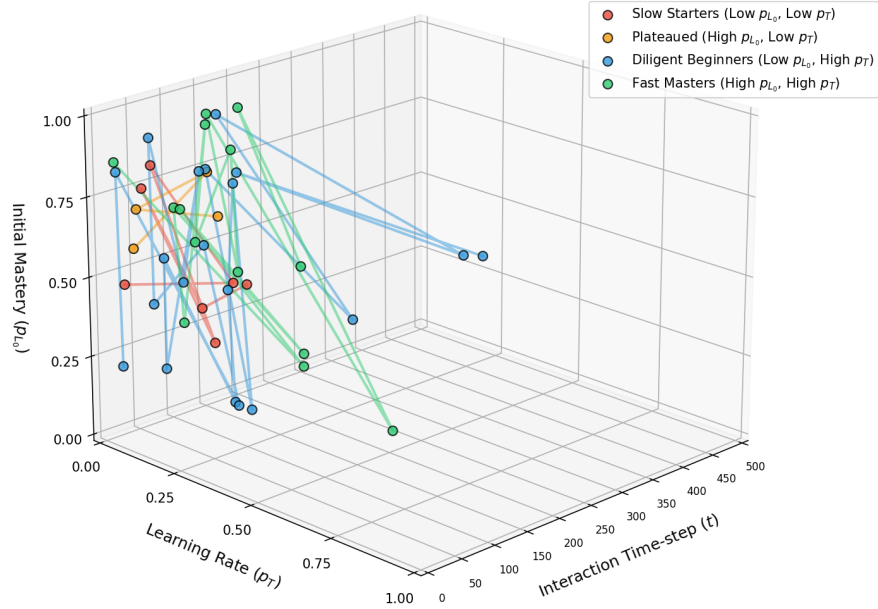
3D Cognitive Roster: Longitudinal Trajectories across Learning Situations

Fig. 1: **3D Cognitive Roster.** Longitudinal trajectories of four representative students across the (p_{L_0}, p_T, t) space. Points represent interactions over time; colours correspond to the four learning situations. Vertical lines indicate interaction time-steps.

declines from 0.077 to 0.050—the system reached a stable high-mastery state and ceased active learning. Diligent Beginners (student 917) exhibit the richest dynamics: both p_{L_0} ($0.52 \rightarrow 0.67$) and p_T ($0.12 \rightarrow 0.20$) increase over time, with points distributed above the diagonal in both panels—an orbit that has not yet converged, indicating ongoing learning. Fast Masters (student 13) show early stabilisation: p_{L_0} is already high from the first interactions (mean 0.74) and changes negligibly thereafter, while p_T declines moderately ($0.15 \rightarrow 0.11$), suggesting the model assigns reducing marginal gains to a student already near mastery.

These results underscore the utility of theory-grounded user modeling for ECTEL 2026’s theme of Mindful TEL. By shaping learning technologies with intention, we ensure that deep learning’s predictive power does not come at the cost of pedagogical transparency. The augmented user model provides the "diagnostic handles" required for educators to exercise agency within AI-mediated

Table 1: **Predictive performance (AUC) comparison.** Bold indicates best performance per dataset.

Model	AS2009	AS2015	AL2005	BDG2006	NIPS34
AKT	0.7825	0.7081	0.8306	0.8208	0.8033
ATKT	0.7715	0.7810	0.7995	0.7889	0.7665
DKT	0.7528	0.7031	0.8149	0.8015	0.7689
DKVMN	0.7440	0.7013	0.8054	0.7983	0.7673
SAKT	0.7263	0.6947	0.7880	0.7740	0.7517
SAINT	0.6921	0.6836	0.7775	0.7781	0.7873
gTransformer	0.7831	0.7078	0.8240	0.8148	0.8006

environments, transforming the "black box" into a collaborative cognitive engine that supports, rather than replaces, human expertise.

6 Experimental Setup

We evaluate gTransformer on five benchmark datasets with explicit knowledge component mappings: ASSISTments 2009 (AS2009, 4,217 students, 123 skills), ASSISTments 2015 (AS2015, 19,917 students, 100 skills), Algebra 2005 (AL2005, 574 students, 112 skills), Bridge to Algebra 2006 (BDG2006, 1,146 students, 493 skills), and NeurIPS 2020 Education Challenge (NIPS34, 4,918 students, 57 skills). We follow a standard five-fold cross-validation with an 80/20 train/test split, measuring performance via AUC using the question-level mean late-fusion protocol [?]. Baselines include DKT [?], DKVMN [?], SAKT [?], SAINT [?], AKT [?], and ATKT [?].

6.1 User Model: Theory-Grounded Knowledge Tracing

To quantify gTransformer’s performance as a user model and the cost of pedagogical grounding, we define three metrics. Let p_{sup1} be ablated (non-grounded) supervised predictions, p_{sup2} grounded supervised predictions, p_{ref} interpretable BKT-based predictions, and p_{bkt} classical BKT predictions:

$$\mathcal{I}_{\text{drop}} = p_{\text{sup1}} - p_{\text{sup2}} \quad (\text{cost of grounding on neural capacity}) \quad (1)$$

$$\mathcal{I}_{\text{cost}} = p_{\text{sup2}} - p_{\text{ref}} \quad (\text{cost of switching to interpretable logic}) \quad (2)$$

$$\mathcal{I}_{\text{gain}} = p_{\text{ref}} - p_{\text{bkt}} \quad (\text{advantage over population-level BKT}) \quad (3)$$

Table ?? shows the comparative performance of gTransformer against baselines in the ablated (non-grounded) configuration. gTransformer achieves the best AUC on AS2009, ranks second on AL2005, BDG2006, and NIPS34, and third on AS2015, demonstrating that the architecture is competitive with the state of the art.

Table 2: **AUC–interpretability trade-offs.** $\mathcal{I}_{\text{drop}}$ = cost of grounding; $\mathcal{I}_{\text{cost}}$ = interpretability cost; $\mathcal{I}_{\text{gain}}$ = gain over BKT.

Dataset	p_{sup1}	p_{sup2}	p_{ref}	p_{bkt}	$\mathcal{I}_{\text{drop}}$	$\mathcal{I}_{\text{cost}}$	$\mathcal{I}_{\text{gain}}$
AS2009	0.7831	0.7814	0.7436	0.6097	0.2%	4.8%	22.0%
AS2015	0.7078	0.7070	0.6940	—	0.1%	1.8%	—
AL2005	0.8240	0.8237	0.7800	0.7215	0.0%	5.3%	8.1%
BDG2006	0.8148	0.8107	0.7810	0.6756	0.5%	3.6%	15.6%
NIPS34	0.8006	0.7987	0.7666	0.5729	0.2%	4.0%	33.8%
Mean	0.7861	0.7843	0.7530	0.6449	0.2%	3.9%	19.9%

Table ?? quantifies the trade-off. The interpretability drop ($\mathcal{I}_{\text{drop}}$) is negligible on average (0.2%), demonstrating that grounding constraints do not compromise the model’s expressive capacity. The interpretability cost ($\mathcal{I}_{\text{cost}}$), i.e., the performance reduction when constraining predictions to BKT logic, averages 3.9% across datasets. Against population-level BKT, interpretable gTransformer predictions show a consistent advantage of 19.9% on average. These results establish that pedagogical interpretability can be achieved at a minimal performance cost.

6.2 Domain Model: Validating Theoretical Grounding

To verify that gTransformer’s internal representations are genuinely grounded in the domain model (BKT), we implement a triple-validation methodology: (H2.1) Structural Alignment, assessed via diagnostic probing [?,?] measuring the linear recoverability of BKT constructs from hidden states; (H2.2) Semantic Alignment, assessed via Spearman’s rank correlation (ρ) between grounded parameters and theoretical priors; and (H2.3) Functional Alignment, assessed via a composite confidence metric quantifying the reliability of interpretable predictions as substitutes for supervised ones. We validate using AS2009 as the reference dataset.

Structural Alignment (H2.1). Using diagnostic probing, we assess whether BKT constructs are linearly recoverable from final hidden states. Initial Mastery achieves fidelity $R^2 = 0.549$ ($r = 0.743$) with selectivity $\Delta R^2 = 0.619$, exceeding the 0.5 threshold that distinguishes genuine structural encoding from spurious correlations. Learn Rate achieves $R^2 = 0.515$ ($\Delta R^2 = 0.570$). These results confirm that BKT constructs are preserved as primary organising principles in the latent space.

Semantic Alignment (H2.2). Initial Mastery shows weaker monotonic alignment with theoretical priors ($\rho = 0.360$, $\text{MAE} = 0.168$), reflecting strong student-specific individualisation. Learn Rate demonstrates moderate alignment ($\rho =$

0.544, MAE = 0.092), preserving the pedagogical understanding of practice effects. Both parameters remain within pedagogical bounds, confirming that the model neither abandons nor mechanically reproduces theoretical priors.

Functional Alignment (H2.3). A composite confidence metric shows that 89.2% of student–skill pairs achieve medium or high confidence in interpretable predictions, indicating that theory-grounded outputs provide actionable diagnostic value.

6.3 Tutorial Model: Situation-Based Instruction

The theory-grounded parameters produced by gTransformer—student-specific estimates of initial mastery (p_{L_0}) and learn rate (p_T)—provide the diagnostic handles through which an ITS tutorial model can implement *situation-based instruction*. Rather than applying a uniform instructional strategy, the tutorial model can route each student to a differentiated instructional pathway based on their learning situation as identified by gTransformer. The following section describes how this routing is operationalised through context-aware personalisation.

7 Context-Aware Personalisation

A key contribution of this work is demonstrating how gTransformer enables the ITS tutorial model to move from population-level strategies towards situation-based, context-aware instruction. We cluster students in AS2009 along two dimensions—initial mastery (p_{L_0}) and learn rate (p_T)—identifying four learning situations: (1) low initial mastery/low learn rate, (2) low initial mastery/high learn rate, (3) high initial mastery/low learn rate, and (4) high initial mastery/high learn rate. Each situation calls for a distinct tutorial strategy: students in situation (1) require targeted scaffolding; those in situation (4) are ready for accelerated pacing; intermediate situations benefit from adjusted challenge levels or motivational support.

Figure ?? shows mastery trajectories for students who provide identical response sequences to the same questions but belong to different learning situations. Classical BKT predictions (dotted grey lines) overlap perfectly, reflecting its Markovian constraint: given the same response sequence, BKT produces identical probability estimates regardless of learning history. gTransformer predictions (solid coloured lines) diverge based on the full interaction history, demonstrating the model’s capacity to differentiate students not by what they answered, but by how they have learned across their entire trajectory.

This divergence has direct implications for the tutorial model. Each student interaction event can be routed to a different instructional pathway based on the learning situation assigned by gTransformer. A student whose trajectory places them in situation (1) triggers a scaffolding strategy; a student in situation (4) is routed towards accelerated content progression. Crucially, this routing is not

static: as new interactions arrive, gTransformer updates its parameter estimates from the full longitudinal context, allowing the tutorial model to dynamically re-assign students across situations as their learning trajectory evolves. This event-driven, context-aware routing is precisely what population-level models such as standard BKT cannot support—two students with an identical recent response pattern may receive different instructional interventions if their prior histories place them in different learning situations.

8 Discussion

8.1 Theory-Grounded Instructional Decisions

The results demonstrate that gTransformer addresses the accuracy–interpretability trade-off in KT by providing pedagogically grounded parameters at a minimal performance cost. This is directly relevant to the ECTEL 2026 theme of Mindful TEL: the model embodies the principle that learning technologies should be shaped with intention, ensuring that the cognitive engine of an ITS is anchored in established educational theory rather than opaque data-driven heuristics.

By operationalising BKT parameters (P_{L_0} , P_T) derived from deep learning context, gTransformer provides educators with the diagnostic handles through which an ITS can perform mindful instruction. Rather than acting as a black box that issues recommendations without justification, the system grounds its decisions in constructs that practitioners recognise and can validate. This aligns with broader EU imperatives for explainable, accountable AI in education [?], and responds directly to the concerns raised by the Artificial Intelligence Act’s designation of educational AI as high-risk.

8.2 Beyond Static Labelling: Context-Aware Adaptation

A central contribution of this work is the shift from population-level labelling to situation-based diagnosis. The results presented in Section ?? demonstrate that gTransformer’s context-aware predictions provide a higher-resolution understanding of the learning journey than population-level models. By encoding each student’s longitudinal history into personalisable parameters, the model acknowledges that students are dynamic learners whose instructional needs evolve continuously.

This enables educators to identify, for instance, students with high prior knowledge but low learn rate—suggesting disengagement or instructional mismatch—and adapt the pacing accordingly before significant learning gaps emerge. Conversely, students with high prior knowledge and high learn rate can be identified for accelerated progression to prevent unnecessary remediation. This intentional design addresses essential concerns about accountability and pedagogical integrity in automated instruction.

8.3 Pedagogical Integrity and Human Agency

The triple-validation methodology (RQ2) establishes that the model’s internal representations genuinely internalise pedagogical theory rather than merely correlating with it. Structural alignment confirms BKT constructs as primary organising principles; semantic alignment confirms that grounded parameters retain their pedagogical semantics; and functional alignment quantifies the confidence with which educators can trust interpretable predictions. Together, these results support the claim that gTransformer is not merely a high-performance prediction system, but a theory-grounded diagnostic tool that can serve as the cognitive foundation of responsible, human-centred ITS.

9 Conclusions

This work demonstrates how gTransformer [?], a Grounded Transformer that bridges deep learning performance and pedagogical interpretability through *representational grounding*, can serve as the cognitive foundation of a theory-grounded ITS. We evaluated its capacity to instantiate the three core ITS components. As a user model, interpretable gTransformer predictions consistently surpass classical BKT with an average gain of 19.9% AUC, at a low interpretability cost of 3.9% relative to black-box architectures. As a domain model, gTransformer internalises BKT constructs as its primary organising principle in latent representations, confirmed by structural, semantic, and functional alignment. As a tutorial model foundation, gTransformer enables context-aware, situation-based instruction by capturing longitudinal learning patterns that Markovian models cannot represent.

As a foundation for TEL, gTransformer demonstrates that pedagogically justifiable student modeling is achievable without sacrificing the predictive power needed for effective adaptive instruction. The *representational grounding* methodology is domain-independent and may be extended to other theoretical frameworks beyond BKT or to other scientific fields where established theoretical models coexist with high-volume empirical data. Future research will investigate the integration of guess and slip parameters for finer-grained personalisation and the generalisation of gTransformer to alternative educational theories.

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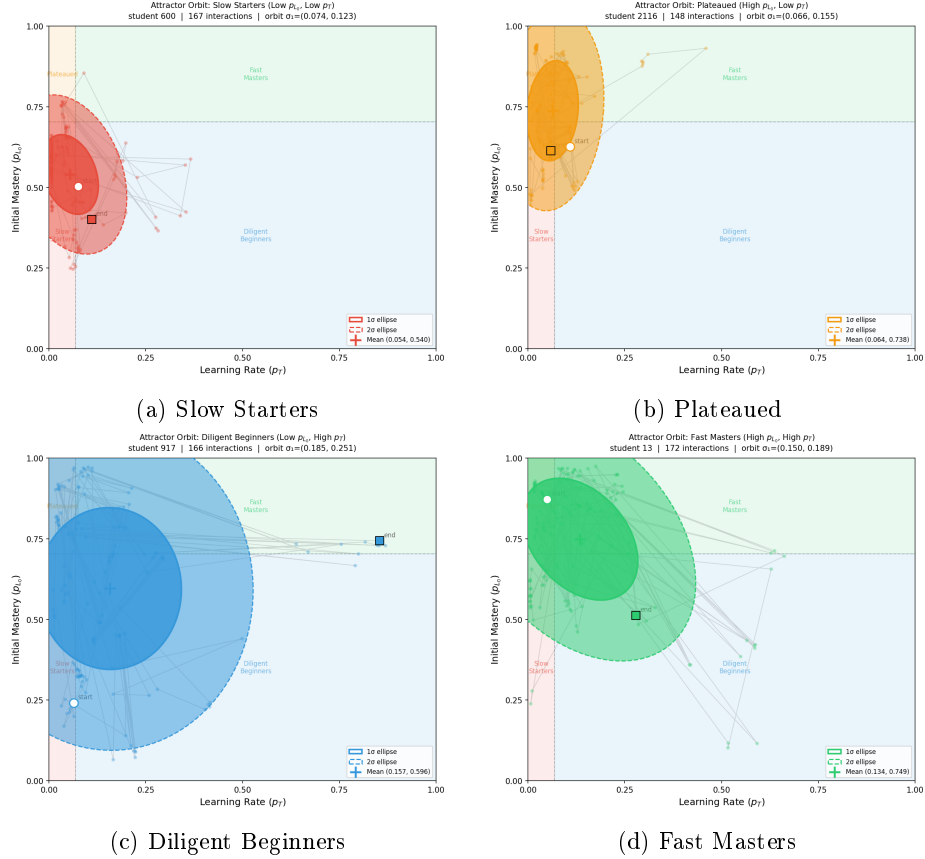


Fig. 2: **Parameter orbit ellipses.** Covariance ellipses (1σ and 2σ) for the representative student of each learning situation in (p_{L0}, p_T) space. The centre marks the mean parameter position; ellipse shape encodes the extent and orientation of parameter variability across all interactions.

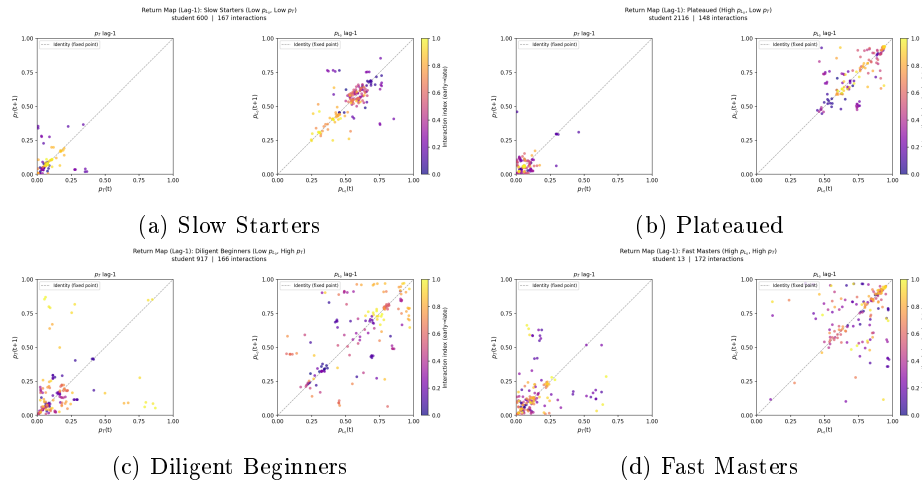


Fig 3: Return maps. Lag plots for the representative student of each learning situation. Each panel shows $p_{L_0}(t+1)$ vs $p_{L_0}(t)$ (left) and $p_T(t+1)$ vs $p_T(t)$ (right). Points are coloured from dark (early) to bright (late) interactions. The diagonal (dashed line) marks the fixed point; deviations above indicate increasing parameters and deviations below indicate decreasing ones.

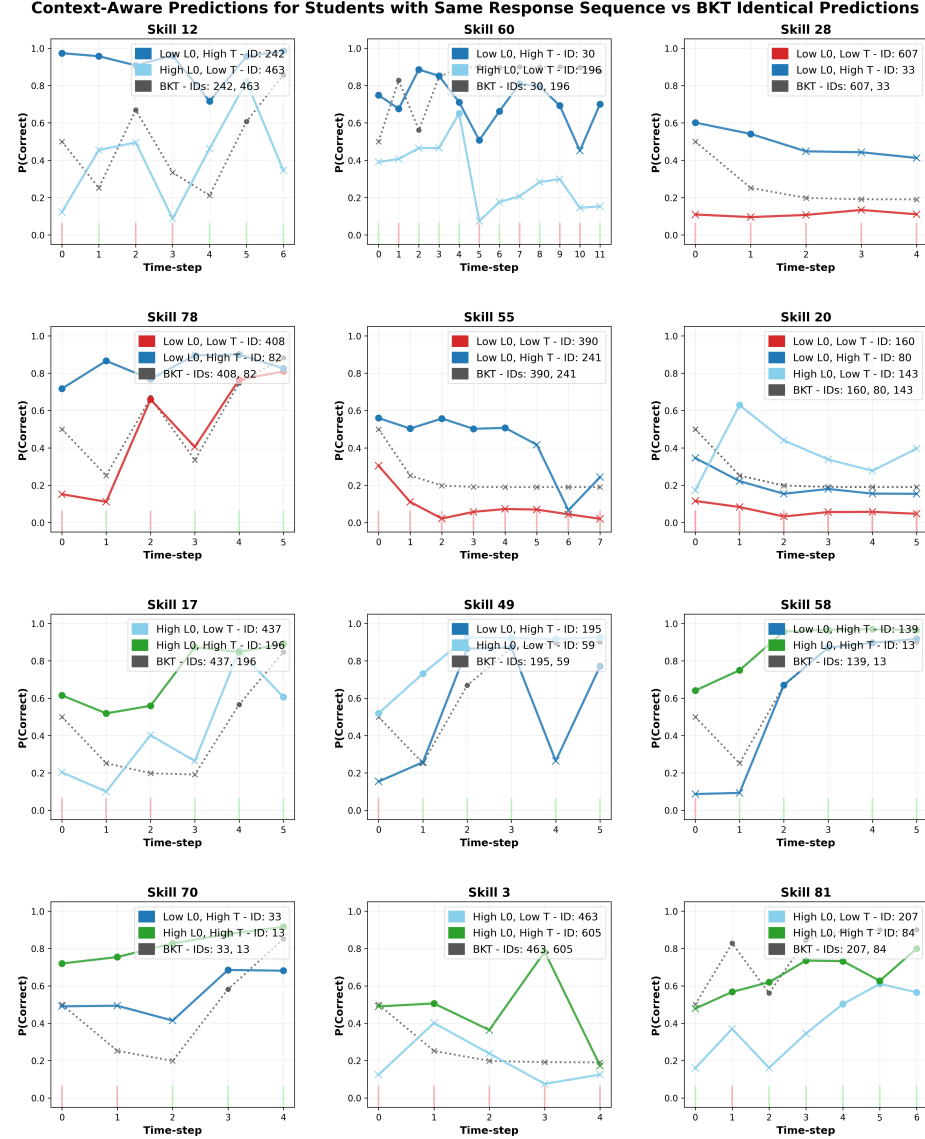


Fig 4: **Context-aware prediction mosaic.** Each panel shows students with identical response sequences (green/red bars for correct/incorrect responses) but different learning situations. BKT predictions (dotted grey) overlap due to Markovian constraints; gTransformer predictions (solid coloured) diverge based on historical parameters (p_{L0} , p_T). Circles indicate predicted correct responses; crosses indicate predicted failures.